Since the Industrial Revolution, humans have been altering Earth's natural environment through increased carbon dioxide (CO_2) emissions. Technological advances—from coal-powered trains, ships, and factories, to oil-fueled cars and planes—have had an adverse impact on the planet, and scientists have formed global research groups to study the phenomenon and suggest ways in which humanity can halt, if not reverse, the damage.

The effects of climate change are varied. As more CO_2 in the atmosphere creates global warming, this causes the polar ice caps to melt, the oceans to warm and rise, and species that are unsuited to warmer oceans to die out. Global weather patterns are also changing: hurricanes in the North Atlantic region have increased in intensity, leaving devastation and death in their wake. Fires and droughts have become more frequent in dry areas; winters are more severe in colder climates. Areas of the world already susceptible to extreme weather-related catastrophes, such as those impacted by tropical monsoons, are seeing the most severe repercussions, especially in terms of loss of life and habitat.